



IMT Atlantique

Bretagne-Pays de la Loire
École Mines-Télécom

Case Study VI: Implementation of Sobel Filter for Edge Detection

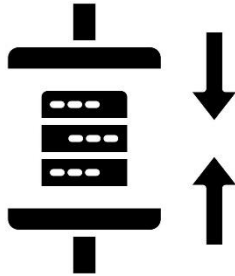
By: Joseph JABOUR

1. INTRODUCTION – EDGE DETECTION

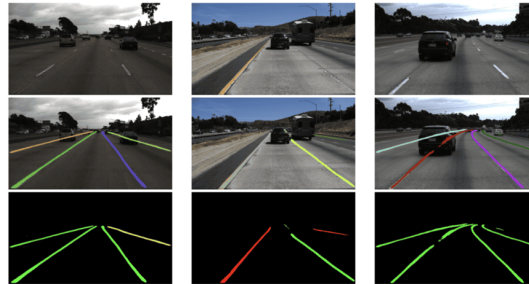
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- Edge detection is fundamental image processing technique, identifies points in image where gradient (rate of intensity change) of image is high.
- These points are typically organized into curved line segments termed "edges"
- Applications:

Data compression



Autonomous Vehicles



Medical Imaging



a)

b)

2. SOBEL FILTER PROCESSING

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- Sobel filter is a high-pass filter used for edge detection of digital images.
- It consists of two kernels K_x and K_y

Horizontal Kernel (K_x)

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical Kernel (K_y)

$$K_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

- The kernels performs two operations:
 1. **Differentiation:** $[-1 \ 0 \ 1]$
 2. **Smoothing:** $[1 \ 2 \ 1]^T$
- These kernels are used to compute the gradient matrices convolutionally.

$$G_x = K_x * A$$

$$G_y = K_y * A$$

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Euclidean Distance

The precise magnitude of the gradient vector.

$$G = \sqrt{G_x^2 + G_y^2}$$

Manhattan Distance

A hardware-friendly approximation avoiding expensive operations.

$$G \approx |G_x| + |G_y|$$

2.1 SOBEL FILTER PROCESSING - EXAMPLE

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$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

195	250	91	23	185	253
35	73	29	231	167	74
95	223	105	83	185	207
38	55	196	108	38	17
55	143	153	16	224	245
54	206	124	64	10	67

	-106				

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	-106	-51	18	40	

3. HARDWARE IMPLEMENTATION

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p_{09}	p_{08}	p_{07}	p_{06}	p_{05}	p_{04}	p_{03}	p_{02}	p_{01}	p_{00}
p_{19}	p_{18}	p_{17}	p_{16}	p_{15}	p_{14}	p_{13}	p_{12}	p_{11}	p_{10}
p_{29}	p_{28}	p_{27}	p_{26}	p_{25}	p_{24}	p_{23}	p_{22}	p_{21}	p_{20}
p_{39}	p_{38}	p_{37}	p_{36}	p_{35}	p_{34}	p_{33}	p_{32}	p_{31}	p_{30}
p_{49}	p_{48}	p_{47}	p_{46}	p_{45}	p_{44}	p_{43}	p_{42}	p_{41}	p_{40}
p_{59}	p_{58}	p_{57}	p_{56}	p_{55}	p_{54}	p_{53}	p_{52}	p_{51}	p_{50}
p_{69}	p_{68}	p_{67}	p_{66}	p_{65}	p_{64}	p_{63}	p_{62}	p_{61}	p_{60}
p_{79}	p_{78}	p_{77}	p_{76}	p_{75}	p_{74}	p_{73}	p_{72}	p_{71}	p_{70}
p_{89}	p_{88}	p_{87}	p_{86}	p_{85}	p_{84}	p_{83}	p_{82}	p_{81}	p_{80}
p_{99}	p_{98}	p_{97}	p_{96}	p_{95}	p_{94}	p_{93}	p_{92}	p_{91}	p_{90}



Gradient
Computation
Block

3. HARDWARE IMPLEMENTATION

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p_{39}	p_{38}	p_{37}	p_{36}	p_{35}	p_{34}	p_{33}	p_{32}	p_{31}	p_{30}
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p_{89}	p_{88}	p_{87}	p_{86}	p_{85}	p_{84}	p_{83}	p_{82}	p_{81}	p_{80}
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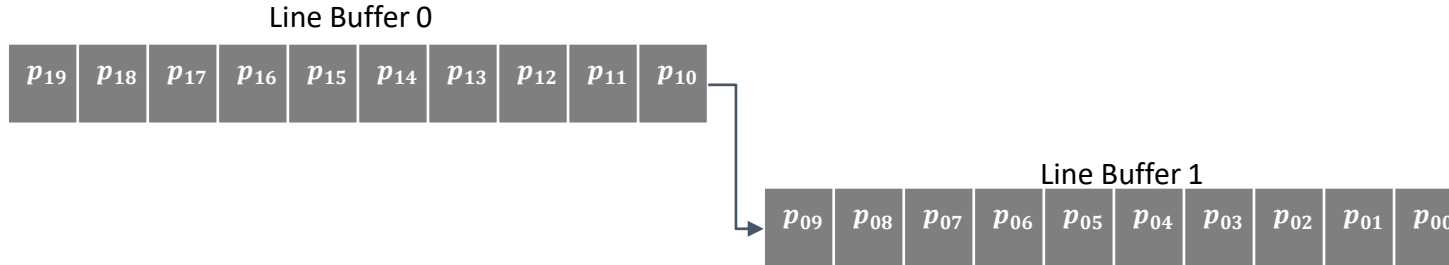
Gradient
Computation
Block

- The system considers serial input.
 - Each clock cycle, 1 pixel is inputted.
- Since pixels arrive serially, we must buffer the previous rows to access a grid of 3×3 pixels.
 - Line Buffer 0 stores first previous row pixels.

Line Buffer 0

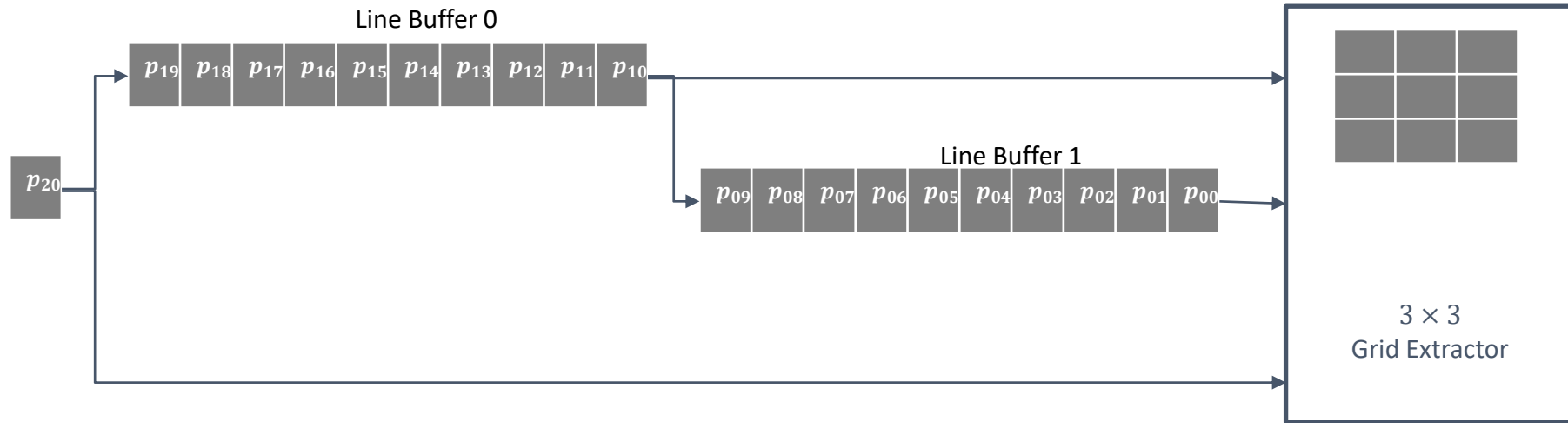


- The system considers serial input.
 - Each clock cycle, 1 pixel is inputted.
- Since pixels arrive serially, we must buffer the previous rows to access a grid of 3×3 pixels.
 - Line Buffer 0 stores first previous row pixels.
 - Line Buffer 1 stores second previous row pixels.

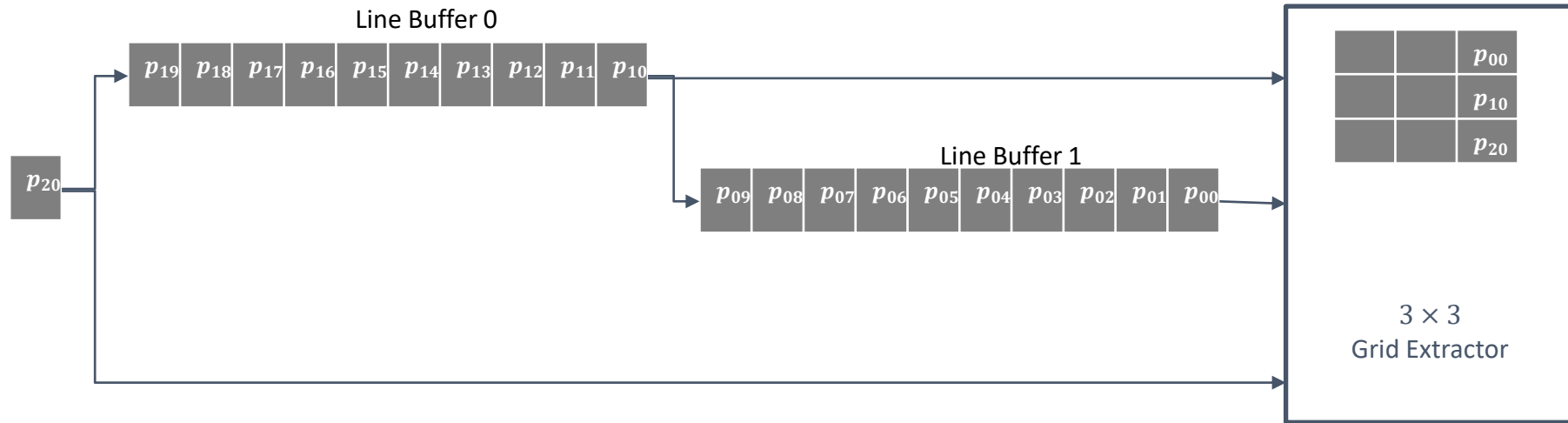


After 20 CCs, Line buffers are filled.

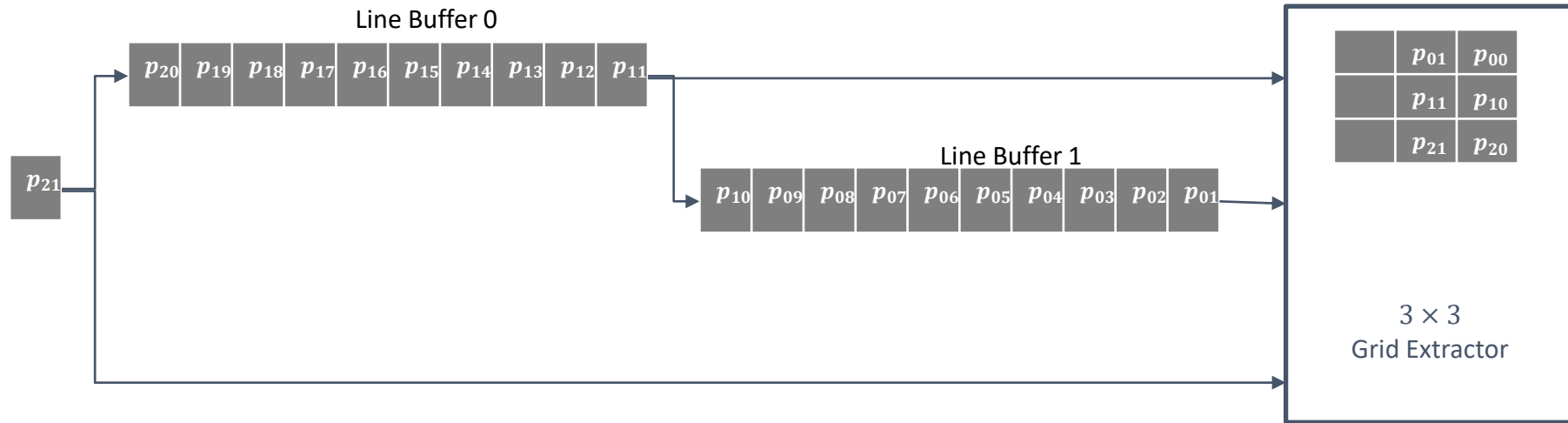
- CC 21



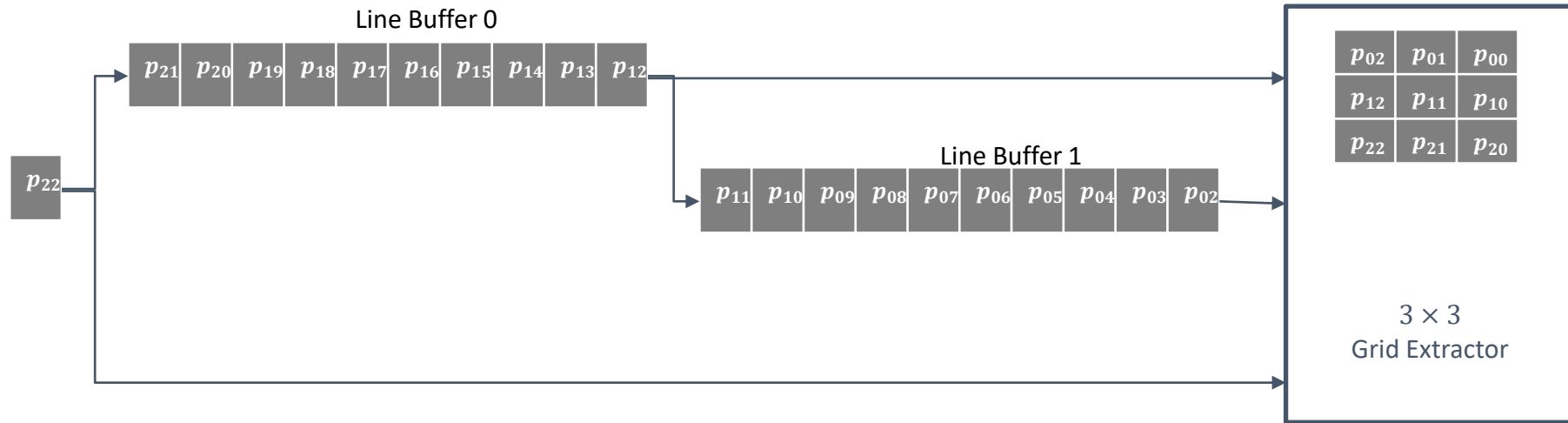
- CC 21



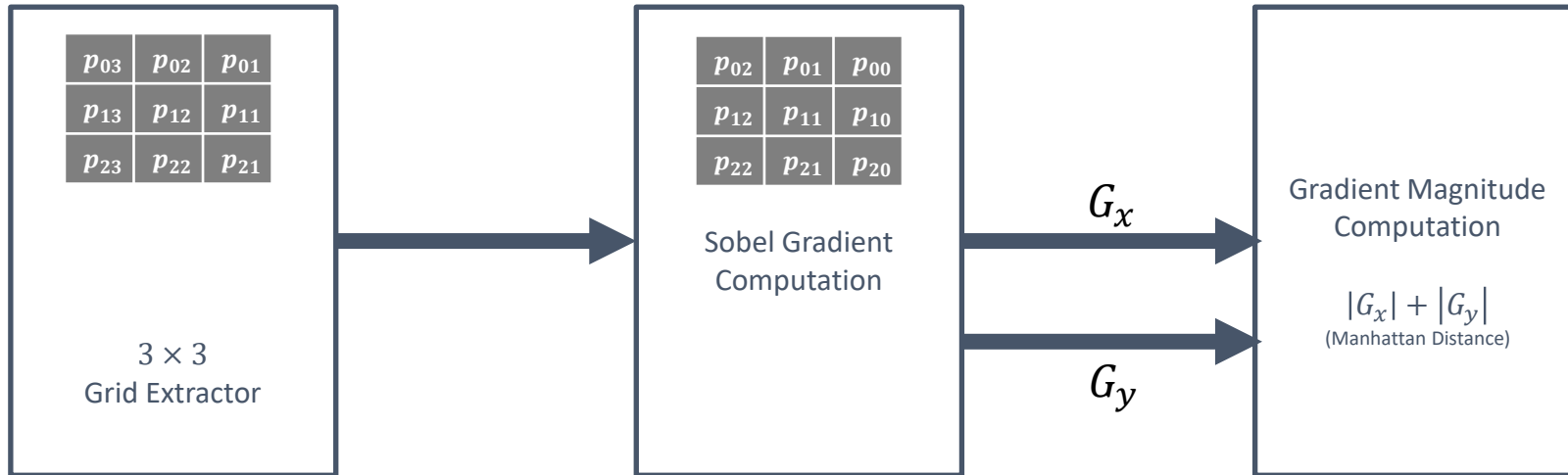
- CC: 22



- CC: 23



- CC: 24



3. SYSTEM ARCHITECTURE

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